



EEE 1101

Electrical Circuits

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Course Content

Basic Circuit Concept:

Basic electrical elements and quantities, Series and parallel circuits, Circuit Laws and Theorems, Mesh and Node analysis, Circuit with dependent sources, Capacitors and Inductors, Fundamental of AC Circuit, Phasor Algebra, AC circuit analysis, Series and parallel resonance, Influence of Circuit Parameters on Resonant Behavior, Q of a circuit, Wave trap, Series-Parallel Tuning.

Balanced Polyphase Circuits:

Two and four phase system, Three-phase four-wire system, Three-phase three-wire system, N-phase star and mesh, Balanced wye and delta loads, Three origin vector diagram of a balanced three phase system, Power measurement (both active and reactive) in balanced systems, Copper required to transmit power under fixed conditions.

Unbalanced Polyphase Circuits:

Unbalanced wye and delta loads, Network solutions, Phase-sequence effects, Methods of checking voltage phase sequence, Power measurements (both active and reactive) in unbalanced three phase systems.



Reference Books

1. Fundamentals of Electric Circuits (5th edition)

- Charles K. Alexander
- Matthew N. O. Sadiku

2. Electric Circuits (10th edition)

- James W. Nilsson
- Susan A. Riedel

3. Alternating Current Circuits (4th edition)

- Russel M. Kerchner
- George F. Corcoran

4. Introductory Circuit Analysis (11th edition)

- Robert L. Boylestad

Polyphase Circuits



Polyphase supply systems have a number of advantages over single-phase systems including the fact that, for a given amount of electrical power transmitted, the total **power loss** is lower in the polyphase system; moreover, the total **volume of conductor** material needed in the cable is less. For these and other reasons, power is transmitted across the nation by a three-phase system.

Another feature is that the torque produced by a single-phase motor is pulsating rather than rotating; that is, an ideal single-phase motor has no **starting torque**! In order to cause the rotor of a single-phase motor to begin to rotate, it is necessary (at starting) to convert it to a two-phase motor. On the other hand, polyphase motors produce a **smooth torque**, and their **speed** can be **controlled** using relatively straightforward methods.

Industry uses many other types of multi-phase system. For example, many control systems use two-phase supplies to drive servomotors, and many rectifier systems use a six-, twelve- or twenty four-phase supply.

Balanced Polyphase Circuits



Generation of Polyphase Voltage:

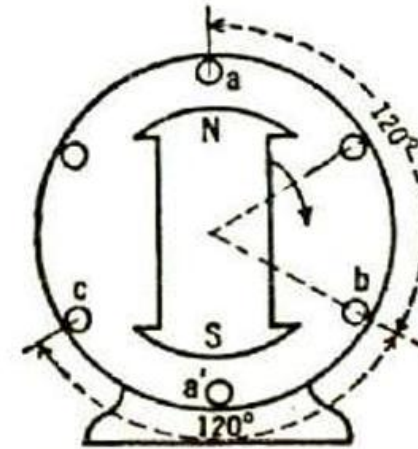
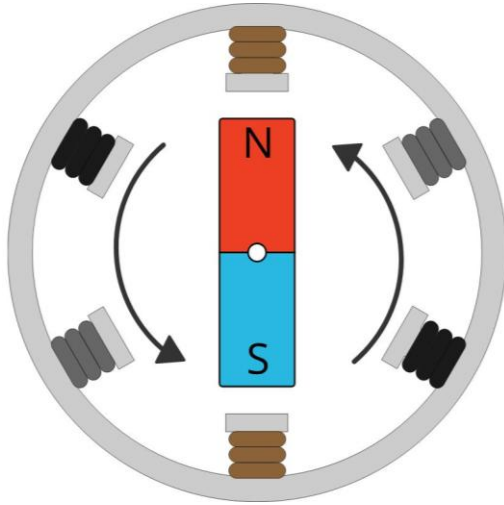
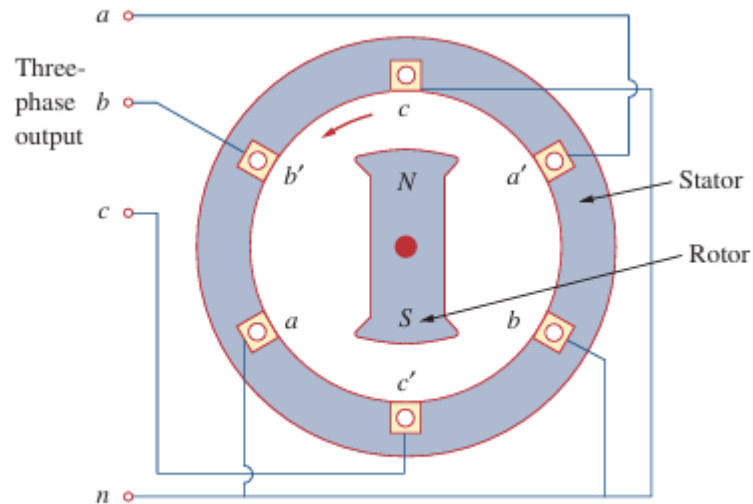


FIG. 1. Elementary three-phase generator.



A three-phase generator.

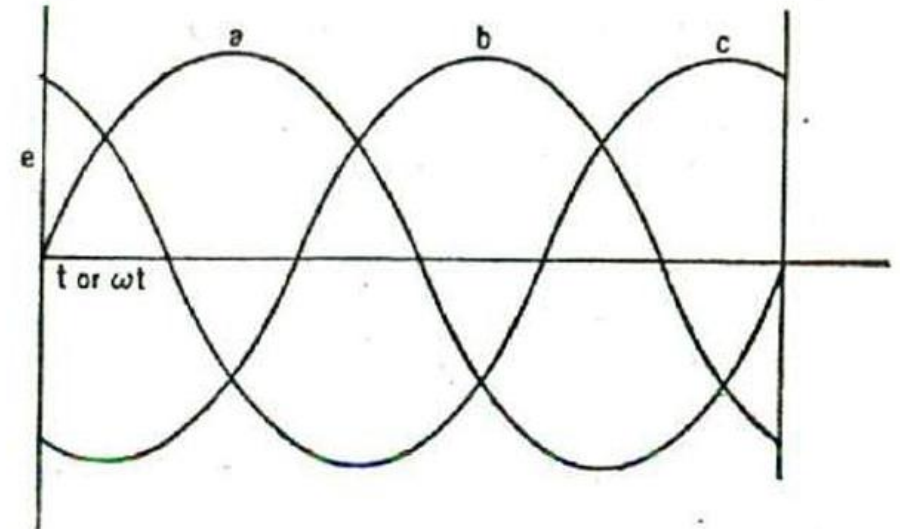


FIG. 3. Waves of emf generated by a three-phase generator.

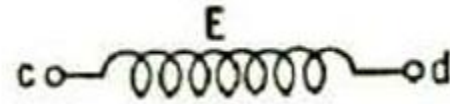
Balanced Polyphase Circuits



Vector Diagram and Double-Subscript Notation:

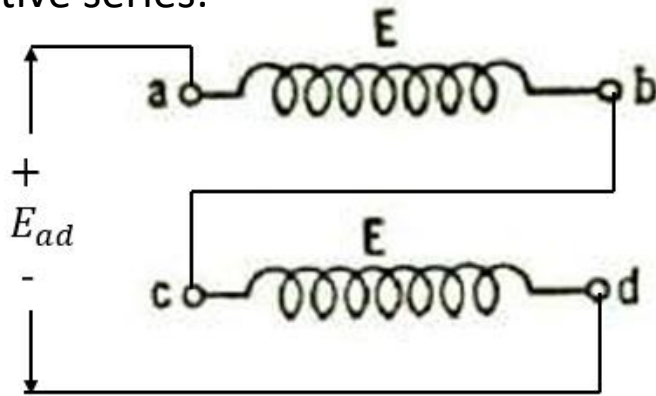


$$E_{ab} = E \angle 0^\circ \text{ V}$$

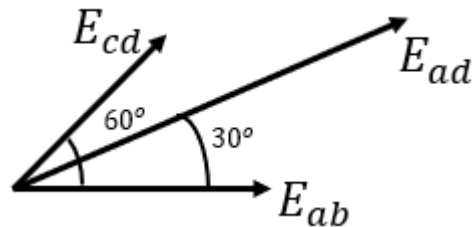


$$E_{cd} = E \angle 60^\circ \text{ V}$$

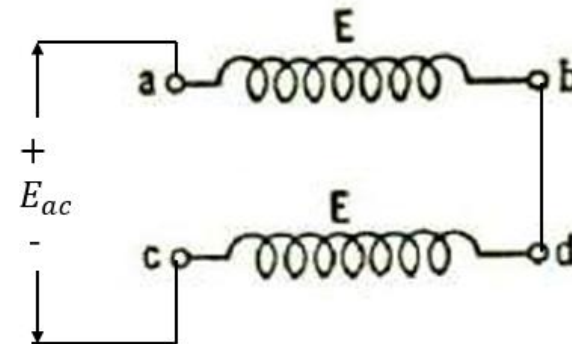
Additive series:



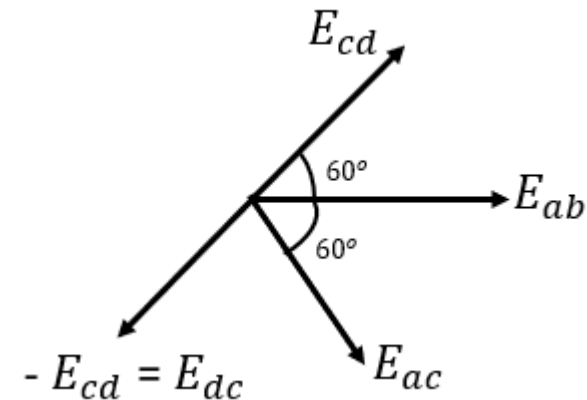
$$E_{ad} = E_{ab} + E_{cd} = E \angle 0^\circ + E \angle 60^\circ = 1.732 E \angle 30^\circ \text{ V}$$



Subtractive series:



$$E_{ac} = E_{ab} - E_{cd} = E \angle 0^\circ - E \angle 60^\circ = E \angle -60^\circ \text{ V}$$



Balanced Polyphase Circuits

Two- and Four-Phase Systems:

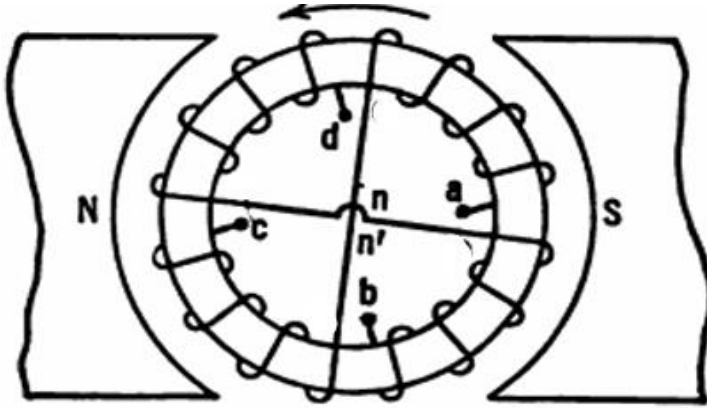


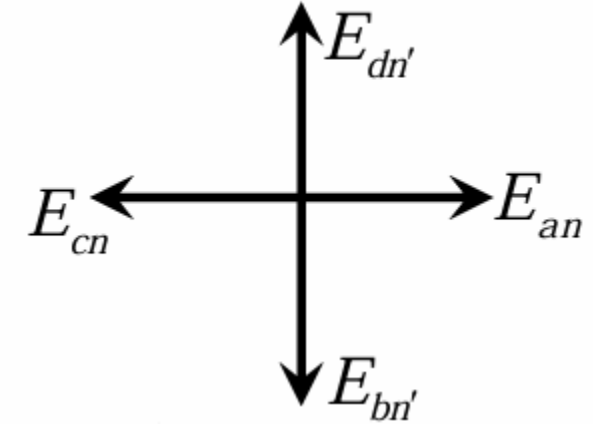
FIG. 8. Elementary Gramme ring-type two-phase generator.

$$E_{an} = E \angle 0^\circ$$

$$E_{bn'} = E \angle -90^\circ$$

$$E_{cn} = E \angle -180^\circ$$

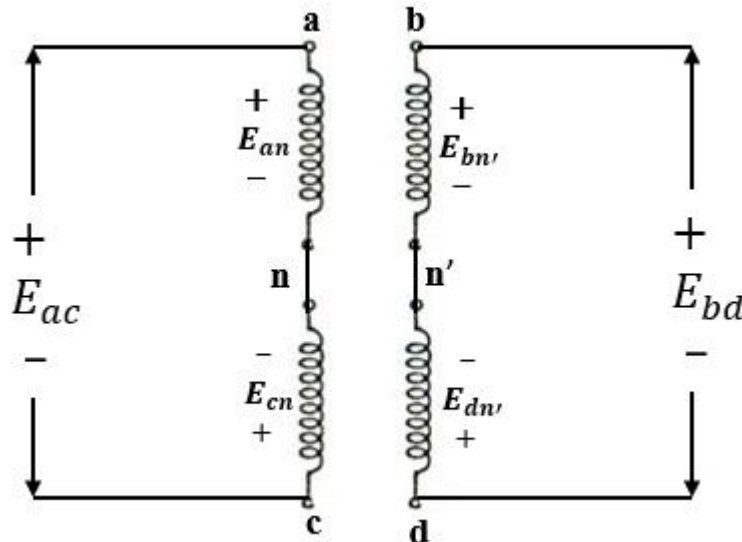
$$E_{dn'} = E \angle -270^\circ$$



1. Two-Phase System:

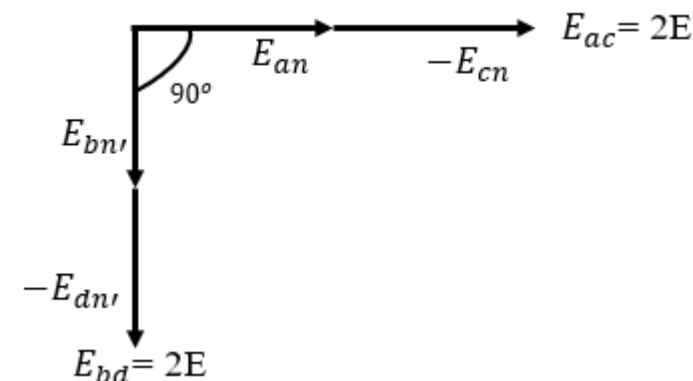
E_{an}

A two-phase system is an electrical system in which the voltages of the phases are 90° out of time phase.



$$E_{ac} = E_{an} - E_{cn} = E \angle 0^\circ - E \angle -180^\circ = 2E \angle 0^\circ \text{ V}$$

$$E_{bd} = E_{bn'} - E_{dn'} = E \angle -90^\circ - E \angle -270^\circ = 2E \angle -90^\circ \text{ V}$$

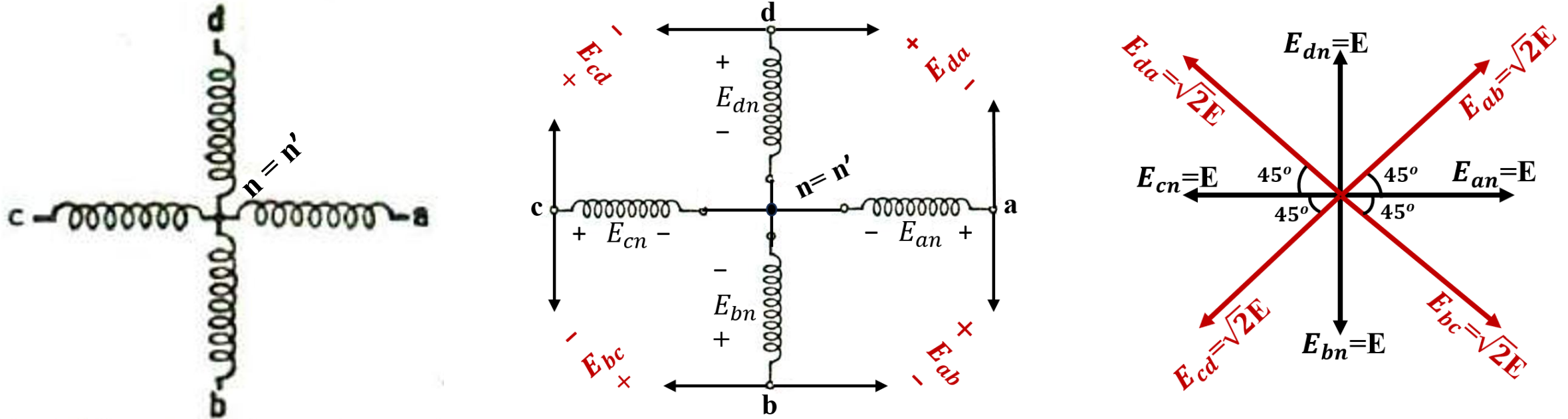


Balanced Polyphase Circuits



2(a). Four-Phase Star System:

If connection is made between the two windings at n and n' , the system would be called a four-phase star system.



Phase Voltages:

$$E_{an} = E \angle 0^\circ$$

$$E_{bn} = E \angle -90^\circ$$

$$E_{cn} = E \angle -180^\circ$$

$$E_{dn} = E \angle -270^\circ$$

Line Voltages:

$$E_{ab} = E_{an} - E_{bn} = E \angle 0^\circ - E \angle -90^\circ = \sqrt{2}E \angle 45^\circ V$$

$$E_{bc} = E_{bn} - E_{cn} = E \angle -90^\circ - E \angle -180^\circ = \sqrt{2}E \angle -45^\circ V$$

$$E_{cd} = E_{cn} - E_{dn} = E \angle -180^\circ - E \angle -270^\circ = \sqrt{2}E \angle -135^\circ V$$

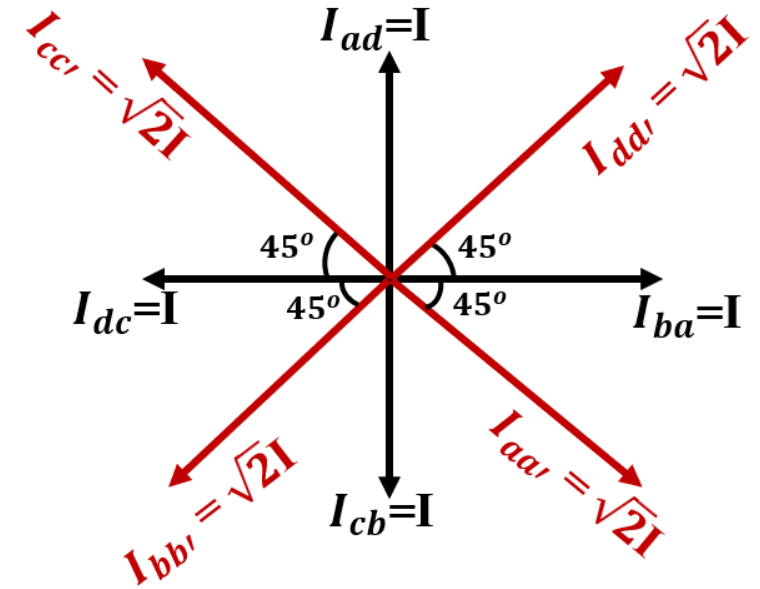
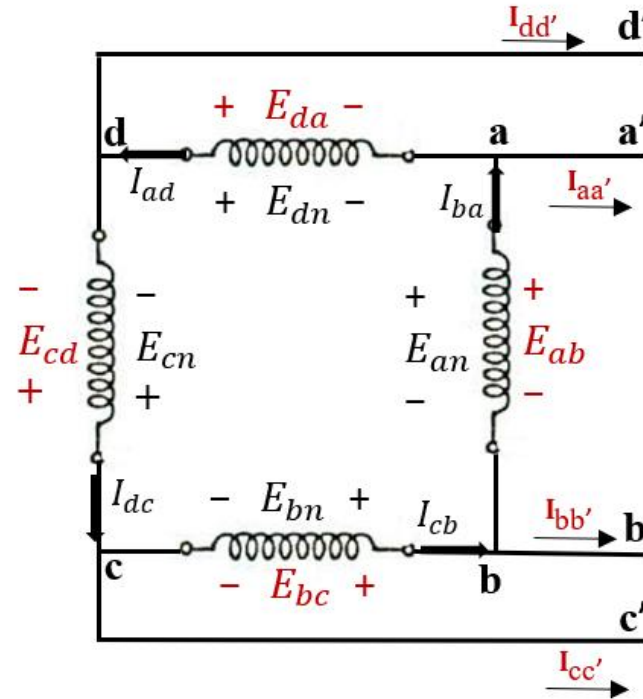
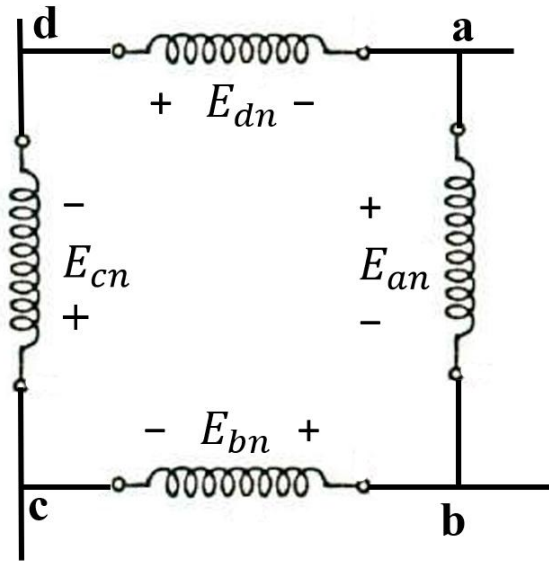
$$E_{da} = E_{dn} - E_{an} = E \angle -270^\circ - E \angle 0^\circ = \sqrt{2}E \angle -225^\circ V = \sqrt{2}E \angle 135^\circ V$$

Balanced Polyphase Circuits



2(b). Four-Phase Mesh System:

Since $E_{an} + E_{bn} + E_{cn} + E_{dn} = 0$, a series connection is possible in a mesh. This configuration is referred to as a four-phase mesh.



Phase Currents:

$$I_{ba} = I \angle 0^\circ$$

$$I_{cb} = I \angle -90^\circ$$

$$I_{dc} = I \angle -180^\circ$$

$$I_{ad} = I \angle -270^\circ$$

Line Currents:

$$I_{aa'} = I_{ba} - I_{da} = I \angle 0^\circ - I \angle -270^\circ = \sqrt{2}I \angle -45^\circ \text{ A}$$

$$I_{bb'} = I_{cb} - I_{ba} = I \angle -90^\circ - I \angle 0^\circ = \sqrt{2}I \angle -135^\circ \text{ A}$$

$$I_{cc'} = I_{dc} - I_{cb} = I \angle -180^\circ - I \angle -90^\circ = \sqrt{2}I \angle 135^\circ \text{ A}$$

$$I_{dd'} = I_{ad} - I_{dc} = I \angle -270^\circ - I \angle -180^\circ = \sqrt{2}I \angle 45^\circ \text{ A}$$

Balanced Polyphase Circuits

Three-Phase, Four-wire Systems (Y-Connected Generator):

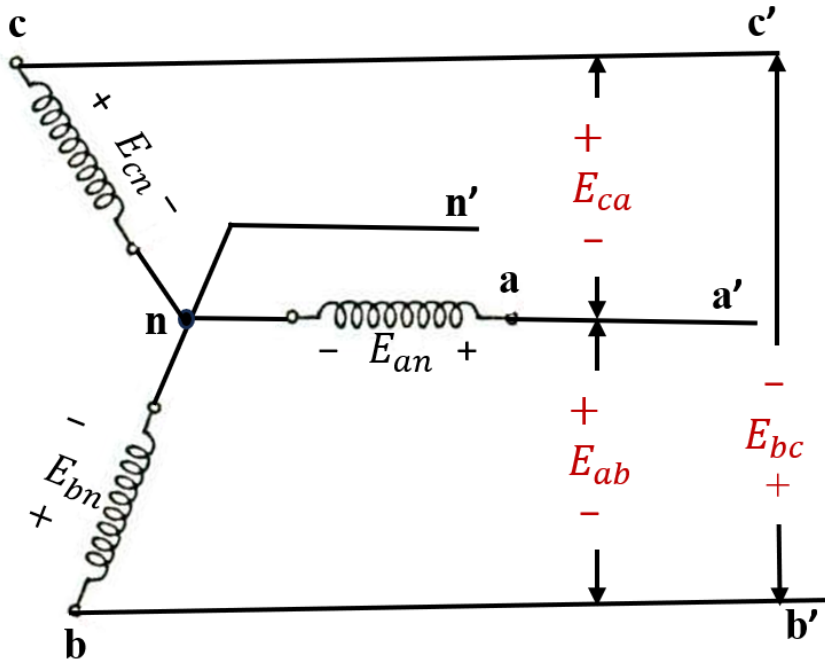


Fig. Three-Phase, Four-wire System (Y-Connection)

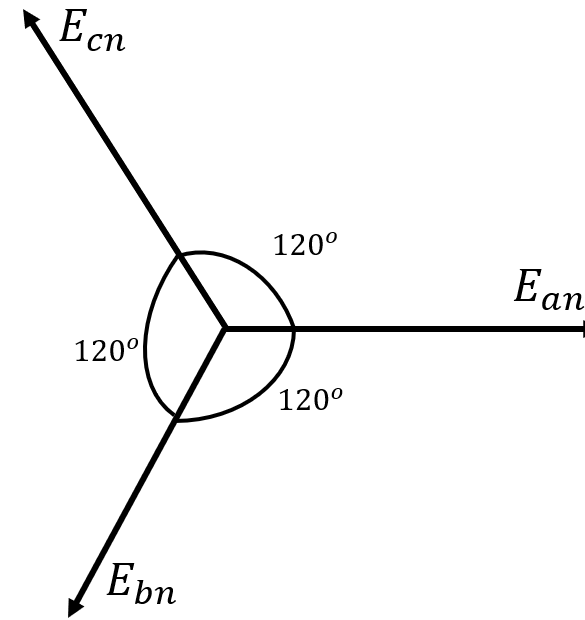


Fig. Phase voltages considering positive phase sequence (a-b-c sequence)

Phase Voltages:

$$E_{an} = E \angle 0^\circ$$

$$E_{bn} = E \angle -120^\circ$$

$$E_{cn} = E \angle -240^\circ = E \angle 120^\circ$$

Line Voltages:

$$E_{ab} = E_{an} - E_{bn} = E \angle 0^\circ - E \angle -120^\circ = \sqrt{3}E \angle 30^\circ \text{ V}$$

$$E_{bc} = E_{bn} - E_{cn} = E \angle -120^\circ - E \angle -240^\circ = \sqrt{3}E \angle -90^\circ \text{ V}$$

$$E_{ca} = E_{cn} - E_{an} = E \angle -240^\circ - E \angle 0^\circ = \sqrt{3}E \angle 150^\circ \text{ V}$$

Balanced Polyphase Circuits

Phase Voltages:

$$E_{an} = E \angle 0^\circ$$

$$E_{bn} = E \angle -120^\circ$$

$$E_{cn} = E \angle -240^\circ = E \angle 120^\circ$$

Line Voltages:

$$E_{ab} = E_{an} - E_{bn} = E \angle 0^\circ - E \angle -120^\circ = \sqrt{3}E \angle 30^\circ \text{ V}$$

$$E_{bc} = E_{bn} - E_{cn} = E \angle -120^\circ - E \angle -240^\circ = \sqrt{3}E \angle -90^\circ \text{ V}$$

$$E_{ca} = E_{cn} - E_{an} = E \angle -240^\circ - E \angle 0^\circ = \sqrt{3}E \angle 150^\circ \text{ V}$$

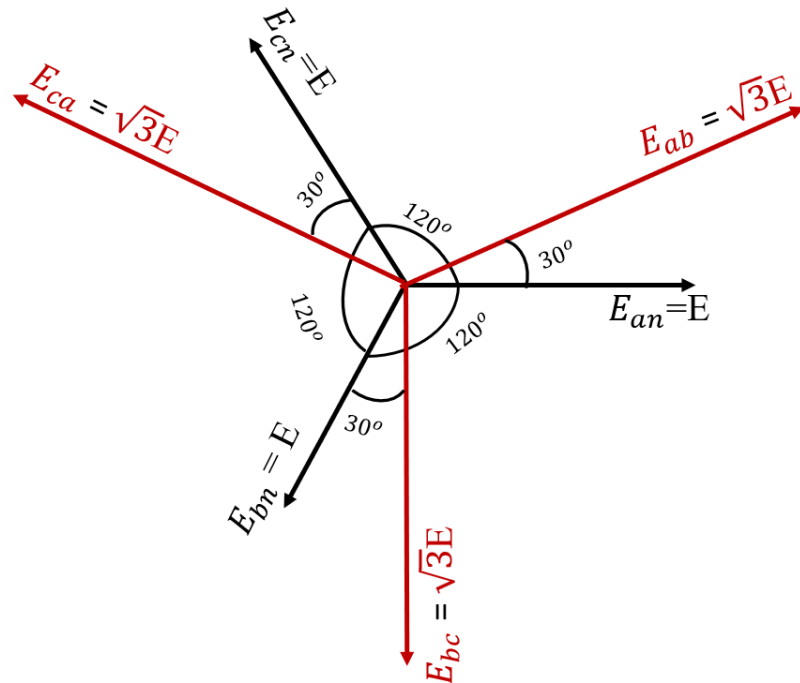


Fig. Line and Phase voltages of the Y-Connection

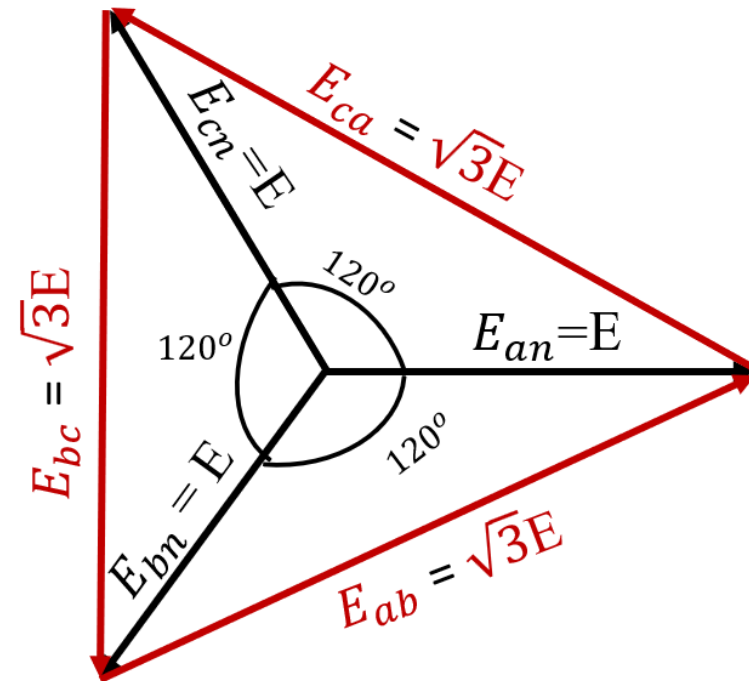
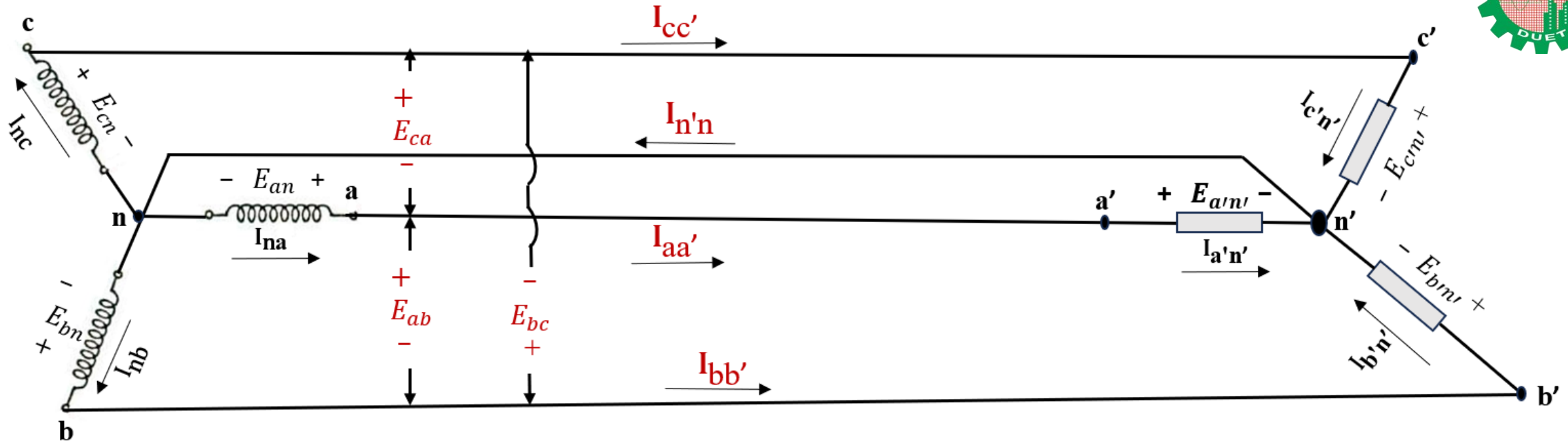


Fig. Vector diagram of Line and Phase voltages for the Y-Connection

In a positive-sequence Y (star) connection, the line voltage is $\sqrt{3}$ times the phase voltage and leads the phase voltage by 30° .

Balanced Polyphase Circuits



Line and phase currents:

$$I_{na} = I_{aa'} = I_{a'n'}$$

$$I_{nb} = I_{bb'} = I_{b'n'}$$

$$I_{nc} = I_{cc'} = I_{c'n'}$$

Therefore, Line current=Phase current

Neutral Currents:

$$I_{n'n} = I_{a'n'} + I_{b'n'} + I_{c'n'}$$

For balanced load,

$$I_{nn'} = 0$$

Phase Voltages of load:

$$E_{a'n'} = E \angle 0^\circ ; E_{b'n'} = E \angle -120^\circ ; E_{c'n'} = E \angle -240^\circ = E \angle 120^\circ$$

Considering Balanced resistive Load, phase currents of load:

$$I_{a'n'} = I \angle 0^\circ ; I_{b'n'} = I \angle -120^\circ ; I_{c'n'} = I \angle -240^\circ = I \angle 120^\circ$$

Substituting these value,

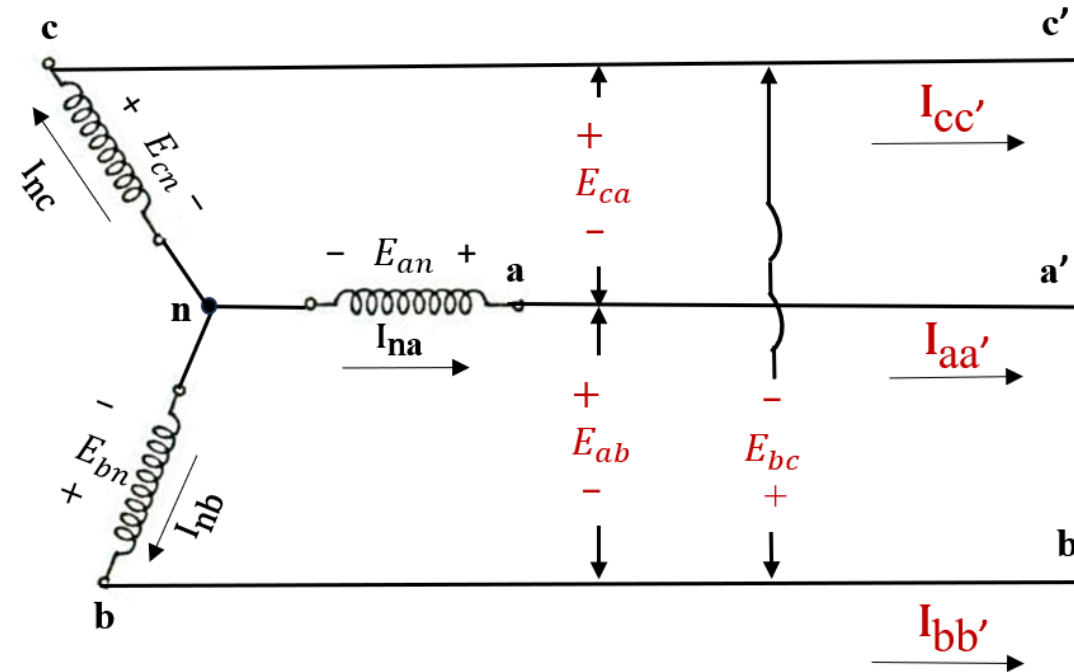
$$\begin{aligned} I_{n'n} &= I_{a'n'} + I_{b'n'} + I_{c'n'} = I \angle 0^\circ + I \angle -120^\circ + I \angle 120^\circ \\ &= I(1 \angle 0^\circ + 1 \angle -120^\circ + 1 \angle 120^\circ) \\ &= 0 \end{aligned}$$

Balanced Polyphase Circuits



Three-Phase, Three-wire Systems (Y-Connection) :

The usual three-phase system consists of only three wires. In this event loads are not placed between the lines and neutral, and the neutral wire is therefore not brought out. The balanced relations discussed in the previous article are obviously unaffected by omitting the neutral wire and therefore apply to the three-phase, three-wire system.



Historical note: Thomas Edison invented a three-wire system, using three wires instead of four.

Balanced Polyphase Circuits

Three-Phase, Three-wire Systems (Δ -Connected Generator):

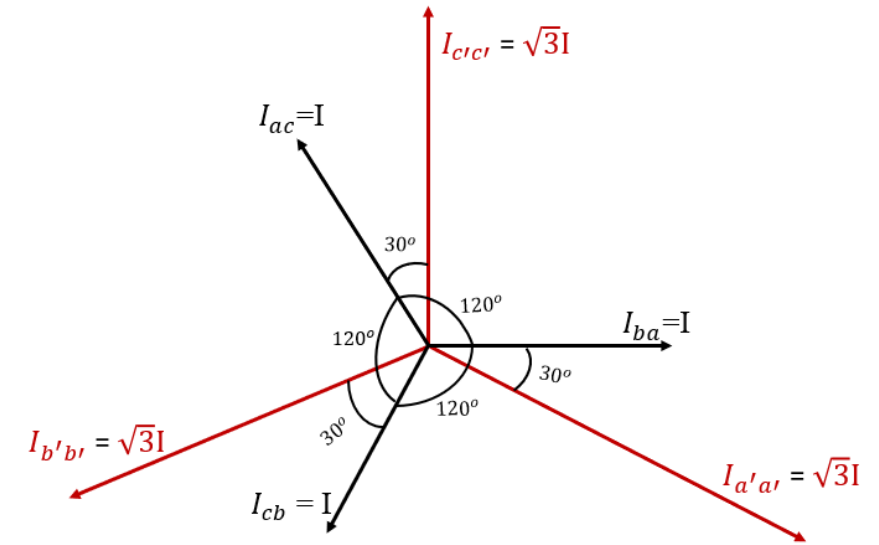
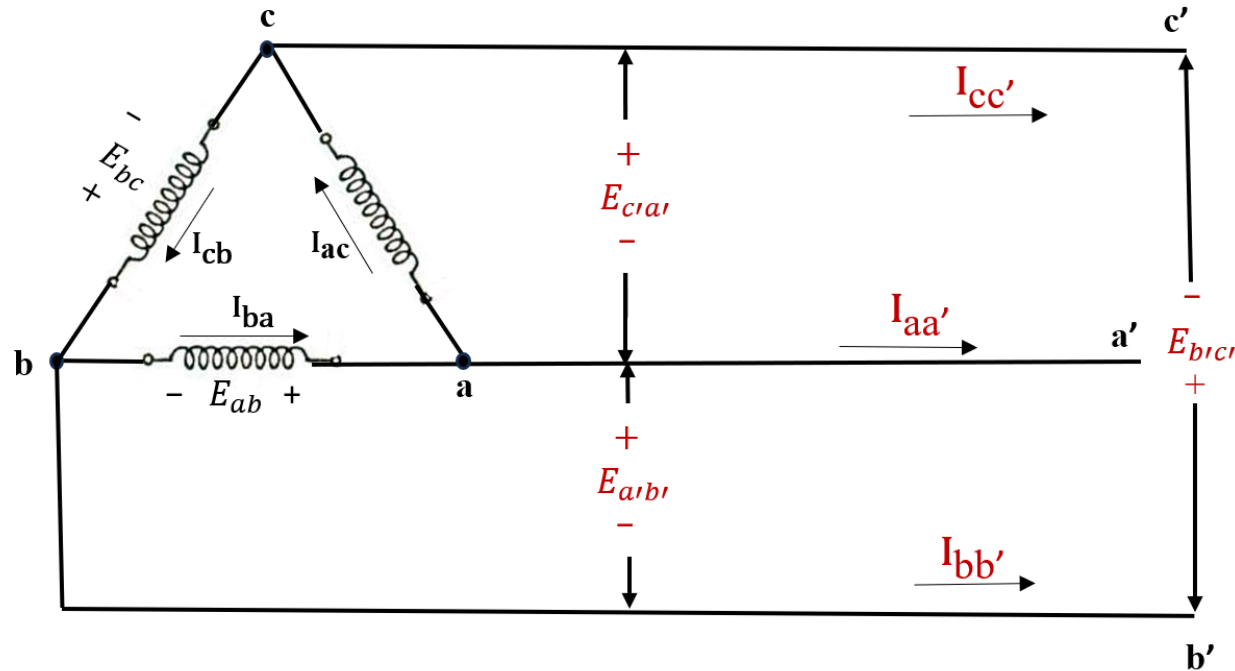


Fig. Line and Phase currents of the Δ -Connected generator

Line and phase Voltages:

$$E_{ab} = E_{a'b'}$$

$$E_{bc} = E_{b'c'}$$

$$E_{ca} = E_{c'a'}$$

Therefore, Line Voltage=Phase Voltage

Phase Currents:

$$I_{ba} = I \angle 0^\circ$$

$$I_{cb} = I \angle -120^\circ$$

$$I_{ac} = I \angle -240^\circ = I \angle 120^\circ$$

Line Currents:

$$I_{aa'} = I_{ba} - I_{ac} = I \angle 0^\circ - I \angle -240^\circ = \sqrt{3}I \angle -30^\circ \text{ A}$$

$$I_{bb'} = I_{cb} - I_{ba} = I \angle -120^\circ - I \angle 0^\circ = \sqrt{3}I \angle -150^\circ \text{ A}$$

$$I_{cc'} = I_{ac} - I_{cb} = I \angle -240^\circ - I \angle -120^\circ = \sqrt{3}I \angle 90^\circ \text{ A}$$

In a positive-sequence delta (Δ)connection, the line current is $\sqrt{3}$ times the phase current and lags the phase current by 30° .

Balanced Polyphase Circuits

Three-Phase, Three-wire Systems (Y-Connected Load):

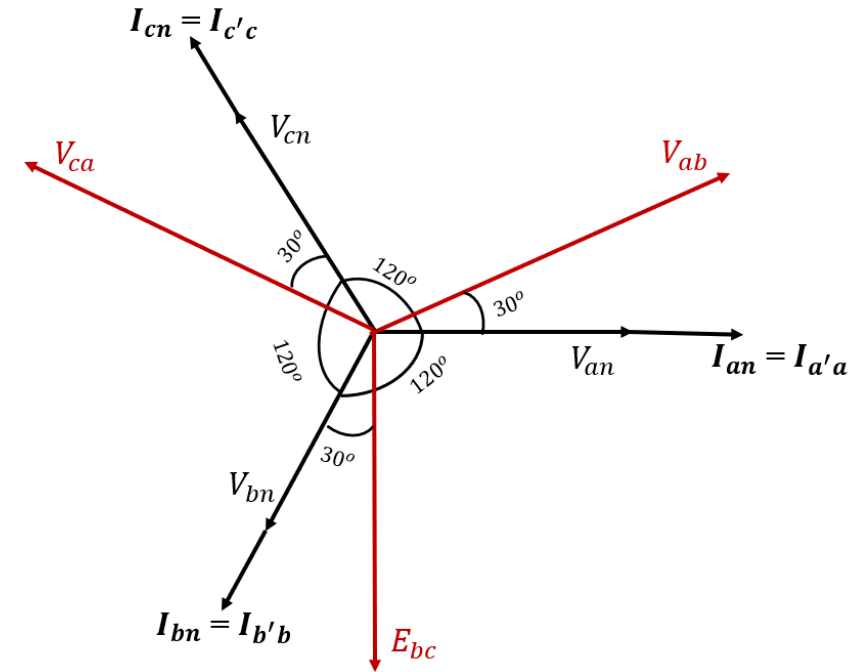
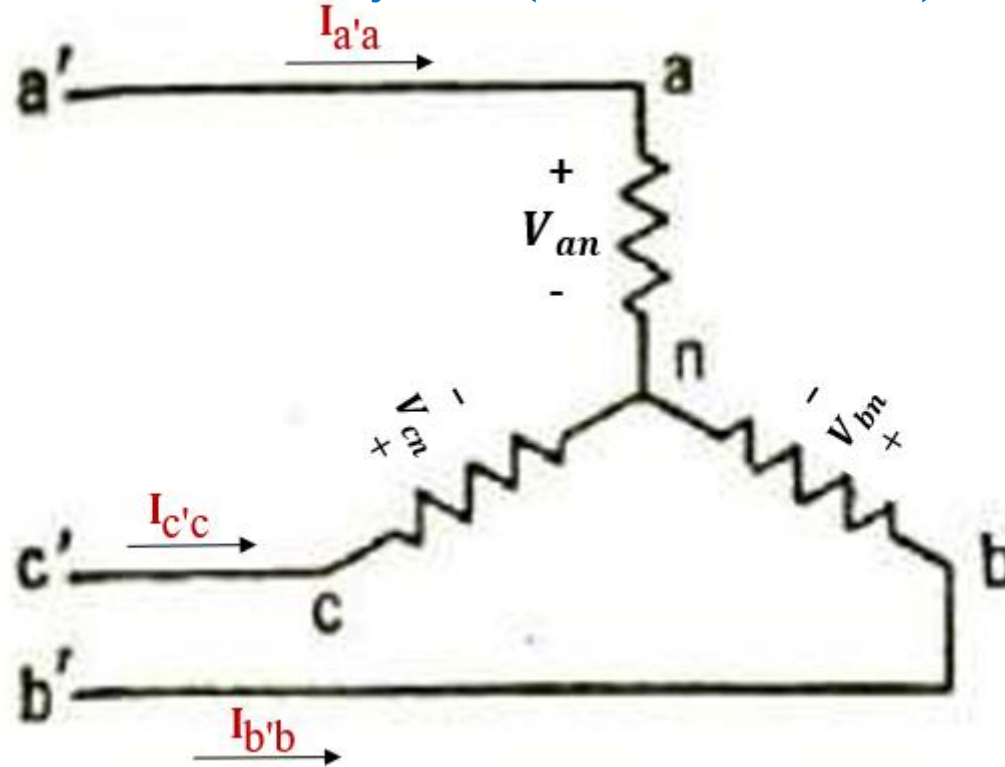


Fig. Vector diagram of the Y-Connected load

Phase Voltages:

$$V_{an} = V \angle 0^\circ$$

$$V_{bn} = V \angle -120^\circ$$

$$V_{cn} = V \angle -240^\circ = V \angle 120^\circ$$

Line Voltages:

$$V_{ab} = V_{an} - V_{bn} = V \angle 0^\circ - V \angle -120^\circ = \sqrt{3}V \angle 30^\circ$$

$$V_{bc} = V_{bn} - V_{cn} = V \angle -120^\circ - V \angle -240^\circ = \sqrt{3}V \angle -90^\circ$$

$$V_{ca} = V_{cn} - V_{an} = V \angle -240^\circ - V \angle 0^\circ = \sqrt{3}V \angle 150^\circ$$

Line and phase currents:

$$I_{an} = I_{a'a}$$

$$I_{bn} = I_{b'b}$$

$$I_{cn} = I_{c'c}$$

Self Study: Draw the Vector diagram of the Y-Connected load for ab-ca-ba sequence.

Balanced Polyphase Circuits

Three-Phase, Three-wire Systems (Δ -Connected Load):

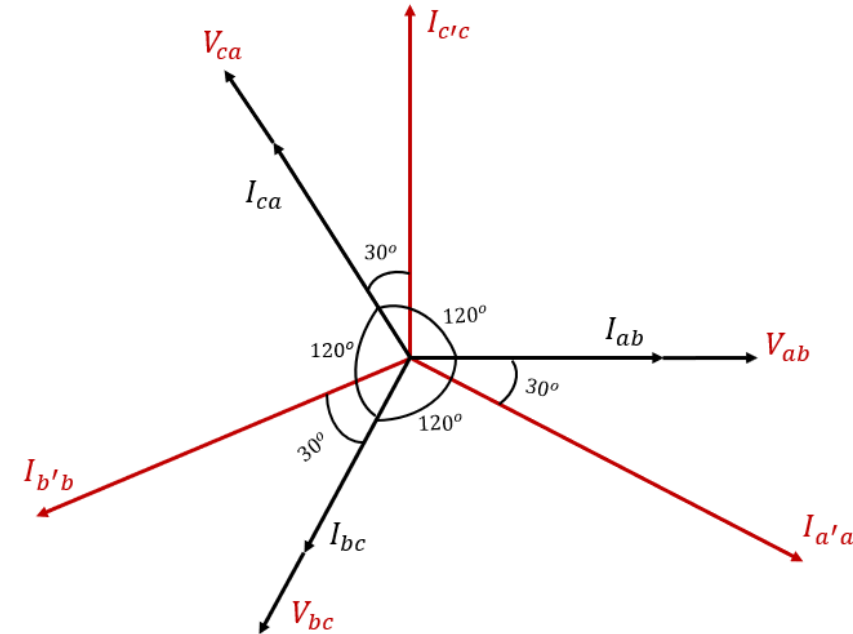
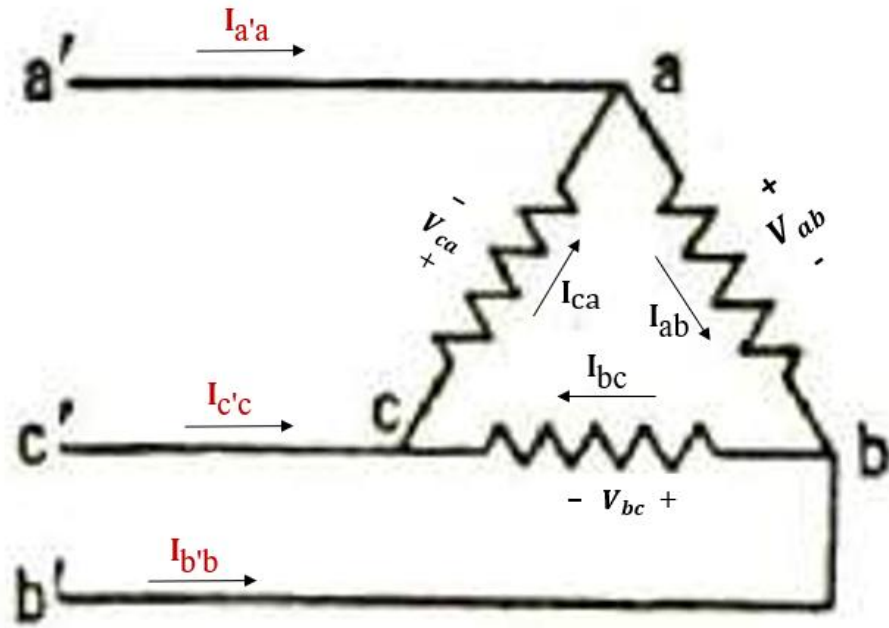


Fig. Vector diagram of the Δ -Connected load

Phase Voltages:

$$V_{ab} = V \angle 0^\circ$$

$$V_{bc} = V \angle -120^\circ$$

$$V_{ca} = V \angle -240^\circ = V \angle 120^\circ$$

Phase Currents:

$$I_{ab} = I \angle 0^\circ$$

$$I_{bc} = I \angle -120^\circ$$

$$I_{ca} = I \angle -240^\circ = I \angle 120^\circ$$

Line Currents:

$$I_{aa'} = I_{ba} - I_{ac} = I \angle 0^\circ - I \angle -240^\circ = \sqrt{3}I \angle -30^\circ \text{ A}$$

$$I_{bb'} = I_{cb} - I_{ba} = I \angle -120^\circ - I \angle 0^\circ = \sqrt{3}I \angle -150^\circ \text{ A}$$

$$I_{cc'} = I_{ac} - I_{cb} = I \angle -240^\circ - I \angle -120^\circ = \sqrt{3}I \angle 90^\circ \text{ A}$$

Self Study: Draw the Vector diagram of the Δ -Connected load for ab-ca-ba sequence

Balanced Polyphase Circuits



Power in Balanced Systems:

If the voltage per phase is V_p , the phase current I_p , and the angle between them θ_p , the power per phase is

$$P_p = V_p I_p \cos \theta_p$$

The power for all phases of an n -phase system is

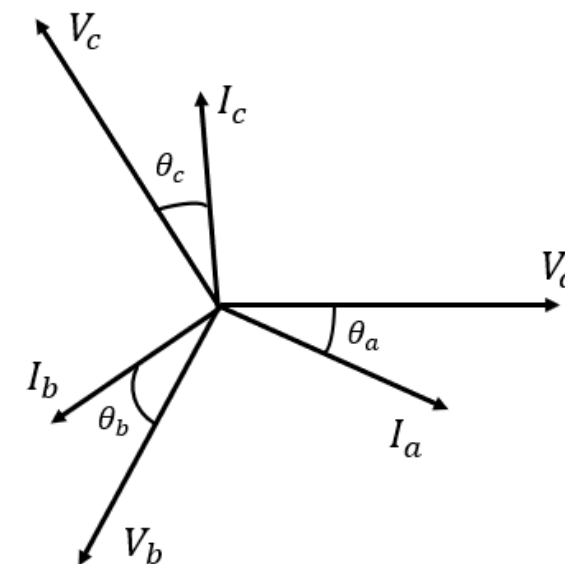
$$P_t = nP_p = nV_p I_p \cos \theta_p$$

The universality of three-phase warrants the development of equation (4) to give power in terms of line current I_L and the line voltage V_L . Consider the wye connection. Then

$$\begin{aligned} P_t &= 3V_p I_p \cos \theta_p = 3 \frac{V_L}{\sqrt{3}} I_L \cos \theta_p \\ &= \sqrt{3} V_L I_L \cos \theta_p \end{aligned}$$

For the delta connection

$$\begin{aligned} P_t &= 3V_p I_p \cos \theta_p = 3V_L \frac{I_L}{\sqrt{3}} \cos \theta_p \\ &= \sqrt{3} V_L I_L \cos \theta_p \end{aligned}$$



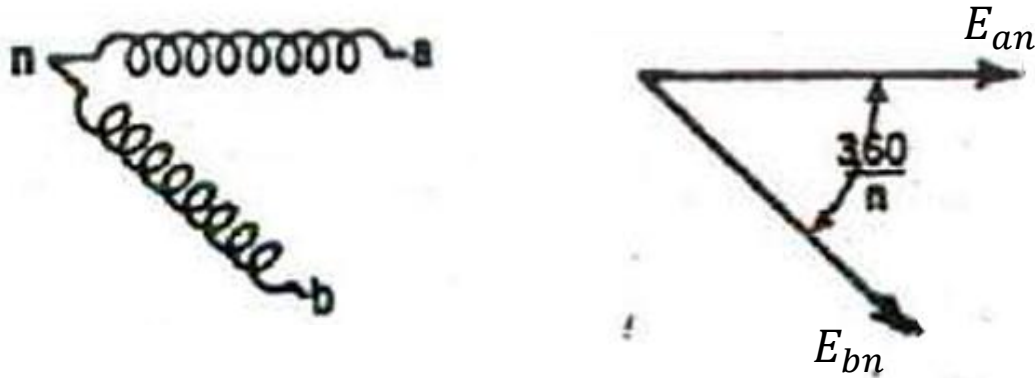
$$\begin{aligned} P &= \sqrt{3} V_L I_L \cos \theta_p \\ P_a &= \sqrt{3} V_L I_L \sin \theta_p \\ P_x &= \sqrt{3} V_L I_L \sin \theta_p \end{aligned}$$

Balanced Polyphase Circuits



n-phase Star and Mesh system:

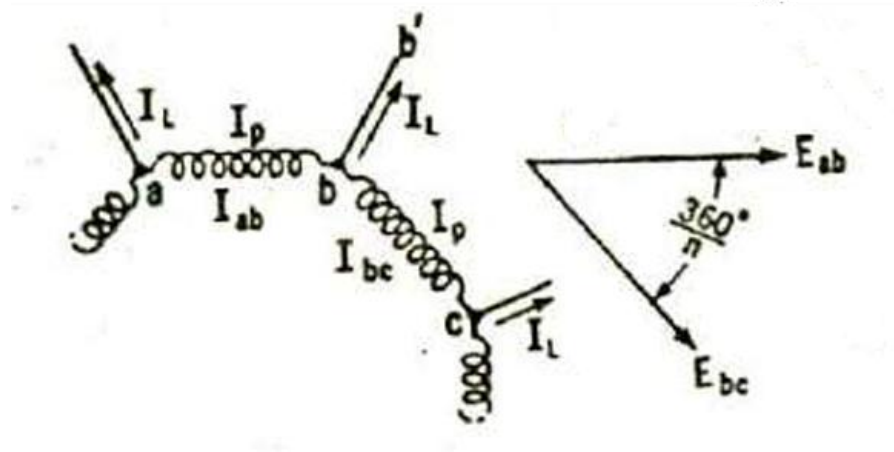
n-phase Star system:



$$E_L = 2E_p \sin \frac{180^\circ}{n}$$

$$I_L = I_p$$

n-phase Mesh system:



$$E_L = E_p$$

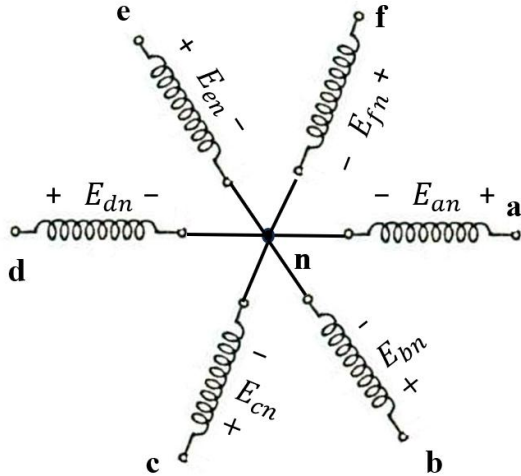
$$I_L = 2I_p \sin \frac{180^\circ}{n}$$

Balanced Polyphase Circuits



6-phase System:

6-phase star system:



Phase Voltages:

$$\begin{aligned} E_{an} &= E \angle 0^\circ \\ E_{bn} &= E \angle -60^\circ \\ E_{cn} &= E \angle -120^\circ \\ E_{dn} &= E \angle -180^\circ \\ E_{en} &= E \angle -240^\circ \\ E_{fn} &= E \angle -300^\circ \end{aligned}$$

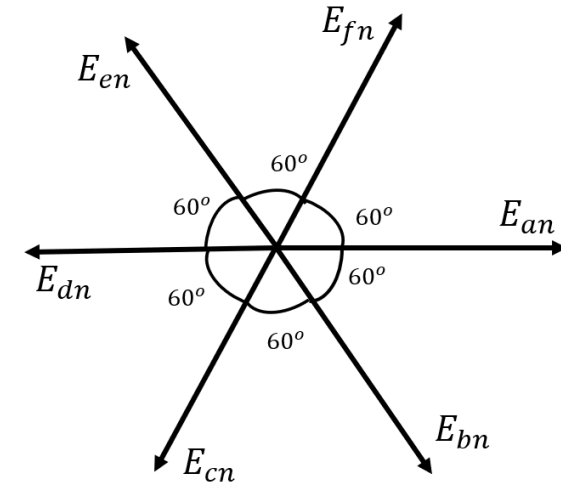
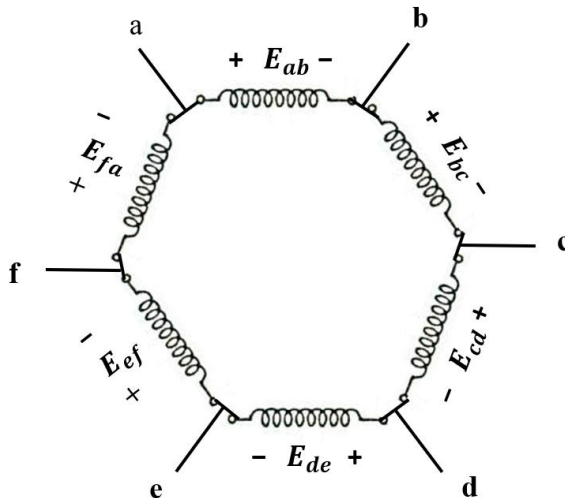


Fig. Vector diagram of the phase voltages of 6-phase star system

6-phase mesh system:



Phase Voltages:

$$\begin{aligned} E_{ab} &= E \angle 0^\circ \\ E_{bc} &= E \angle -60^\circ \\ E_{cd} &= E \angle -120^\circ \\ E_{de} &= E \angle -180^\circ \\ E_{ef} &= E \angle -240^\circ \\ E_{fa} &= E \angle -300^\circ \end{aligned}$$

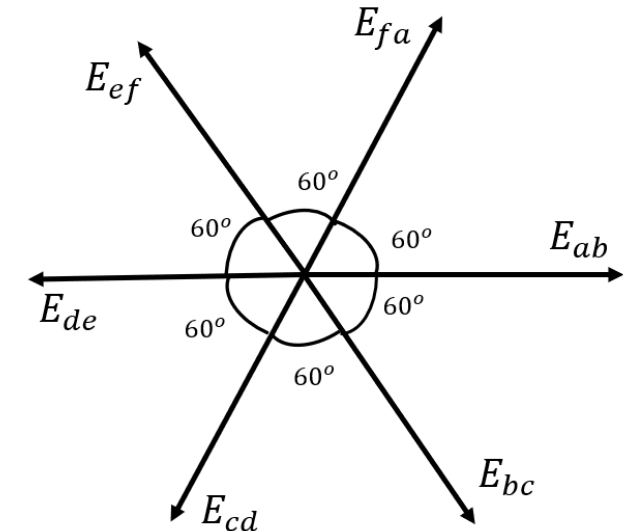


Fig. Vector diagram of the phase voltages of 6-phase mesh system

Balanced Polyphase Circuits

Instantaneous power of three phase system:



$$V_a = V_m \sin \omega t$$

$$V_b = V_m \sin (\omega t - 120^\circ)$$

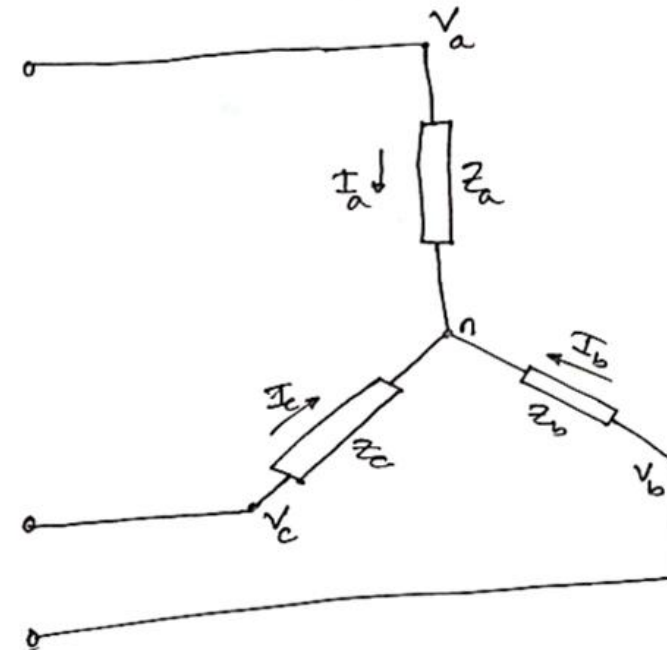
$$V_c = V_m \sin (\omega t - 240^\circ) \\ = V_m \sin (\omega t + 120^\circ)$$

$$Z_a = Z_b = Z_c = Z \angle \theta_p$$

$$i_a = \frac{V_a}{Z_a} = \frac{V_m}{Z} \sin (\omega t - \theta_p) \\ = I_m \sin (\omega t - \theta_p)$$

$$i_b = I_m \sin (\omega t - 120^\circ - \theta_p)$$

$$i_c = I_m \sin (\omega t + 120^\circ - \theta_p)$$



$$P_a = V_a i_a$$

$$P_b = V_b i_b$$

$$P_c = V_c i_c$$

Balanced Polyphase Circuits



$$p_a = V_m I_m \sin \omega t \sin (\omega t - \theta)$$

$$p_b = V_m I_m \sin (\omega t - 120^\circ) \sin (\omega t - 120^\circ - \theta)$$

$$p_c = V_m I_m \sin (\omega t - 240^\circ) \sin (\omega t - 240^\circ - \theta)$$

The total three-phase power is

$$\begin{aligned} p_3 &= P_a + P_b + P_c = V_m I_m [\sin \omega t \sin (\omega t - \theta) \\ &\quad + \sin (\omega t - 120^\circ) \sin (\omega t - 120^\circ - \theta) \\ &\quad + \sin (\omega t - 240^\circ) \sin (\omega t - 240^\circ - \theta)] \end{aligned}$$

$$p_3 = 1.5 V_m I_m \cos \theta$$

Therefore, the instantaneous power of three phase system is independent of time.

Balanced Polyphase Circuits

Power measurement in balanced three phase system:

$$W_a = VI \cos (\theta - 30^\circ)$$

$$W_b = VI \cos (\theta + 30^\circ)$$

$$\begin{aligned} W_a + W_b &= VI \cos (\theta - 30^\circ) + VI \cos (\theta + 30^\circ) \\ &= VI [\cos \theta \cos 30^\circ + \sin \theta \sin 30^\circ + \cos \theta \cos 30^\circ - \sin \theta \sin 30^\circ] \\ &= \sqrt{3} VI \cos \theta \end{aligned}$$

$$\tan \theta = \frac{\sqrt{3} (W_a - W_b)}{W_a + W_b}$$

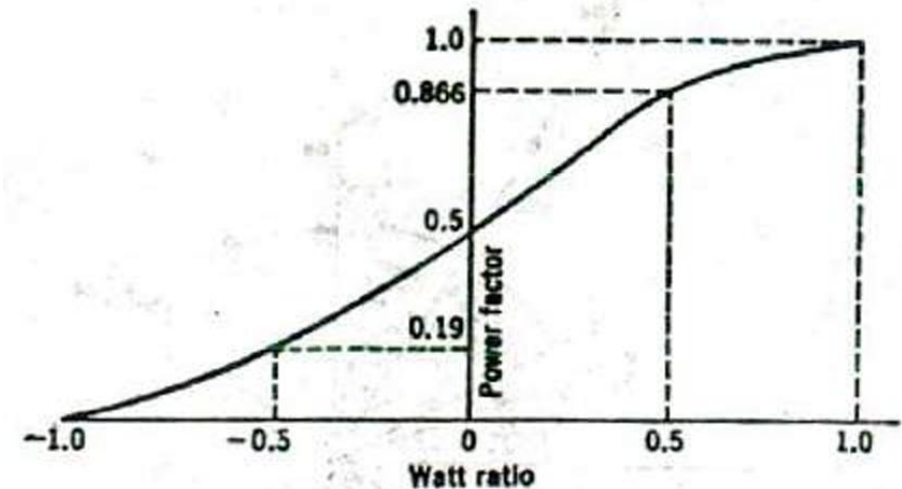
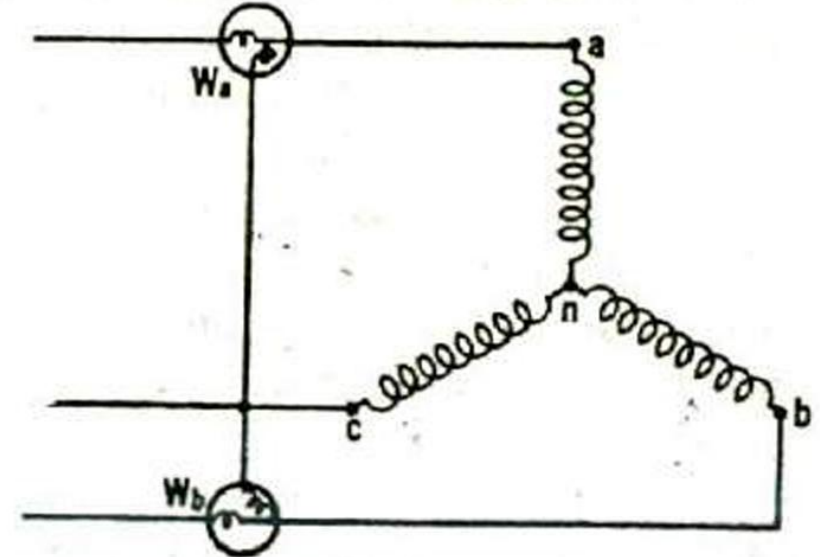


Fig. Watt ratio curve for two-wattmeter method of measuring power (applicable only to balanced loads).



Balanced Polyphase Circuits

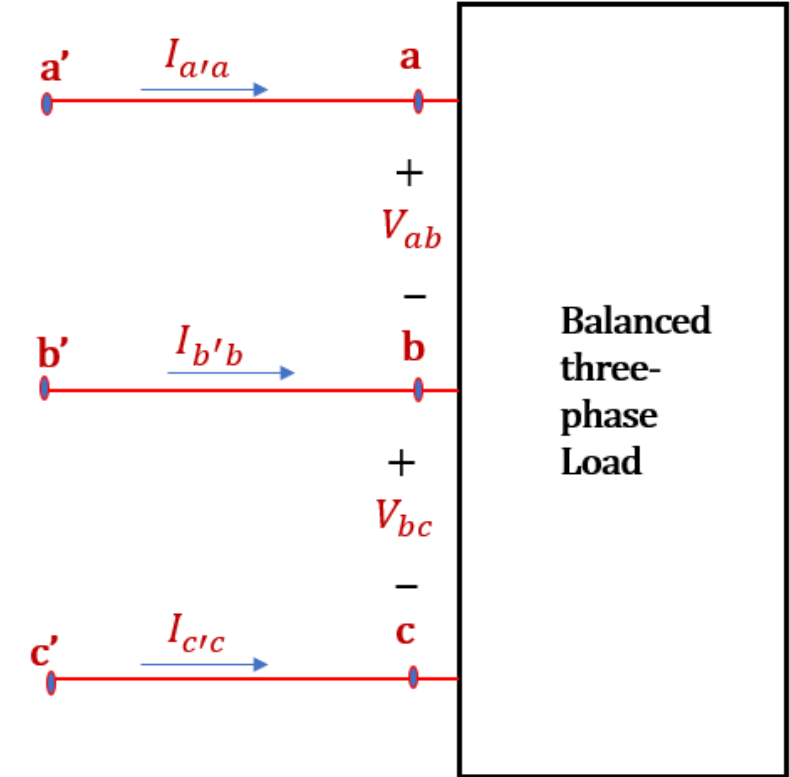


For a balanced 3-phase load,

Line Voltages: $V_{ab} = 400\angle 0^\circ \text{V}$; $V_{bc} = 400\angle -120^\circ \text{V}$; $V_{ca} = 400\angle 120^\circ \text{V}$

Line Currents: $I_{a'a} = 5\angle -30^\circ \text{A}$; $I_{b'b} = 5\angle -150^\circ \text{A}$; $I_{c'c} = 5\angle 90^\circ \text{A}$

Find the power and power factor of the load shown in figure.



Balanced Polyphase Circuits



30. A three-phase, 5-hp, 220-volt induction motor (balanced load) has an efficiency of 86 per cent and operates at 86.6 percent lagging power factor. It is paralleled with a three-phase resistance furnace consisting of three 36-ohm resistances connected in delta. Find the kilovolt-amperes demanded by the combination, the power factor, and the line current.

34. Three $15\angle 60^\circ \Omega$ load impedances are connected in delta and supplied by lines, each line containing 1 ohm resistance and 1 ohm inductive reactance. If the line voltages on the supply side of the line impedances are balanced three-phase of 115 volts each, find the voltage across the load impedances. Also calculate the power loss in the supply lines and the power dissipated by the load itself.

Homework:

From: *Fundamentals of Electric Circuits (5th edition)*

- Chapter-12: Related examples and exercise problems

From: *Alternating-Current Circuit (4th edition)*

- Chapter-VIII : Related examples and exercise problems

Balanced Polyphase Circuits

30. A three-phase, 5-hp, 220-volt induction motor (balanced load) has an efficiency of 86 per cent and operates at 86.6 percent lagging power factor. It is paralleled with a three-phase resistance furnace consisting of three 36-ohm resistances connected in delta. Find the kilovolt-amperes demanded by the combination, the power factor, and the line current.

Solution:

Let the phase sequence is ab-bc-ca.

Line Voltages:

$$V_{ab} = 220\angle 0^\circ \text{V}; V_{bc} = 220\angle -120^\circ \text{V}; V_{ca} = 220\angle 120^\circ \text{V}$$

Phase voltage:

$$V_{an} = V_{AN} = 127\angle -30^\circ \text{V}$$

Input power of motor,

$$P_{in} = \frac{P_{out}}{\eta} = \frac{5 \times 746}{0.86} = 4337.2 \text{ W}$$

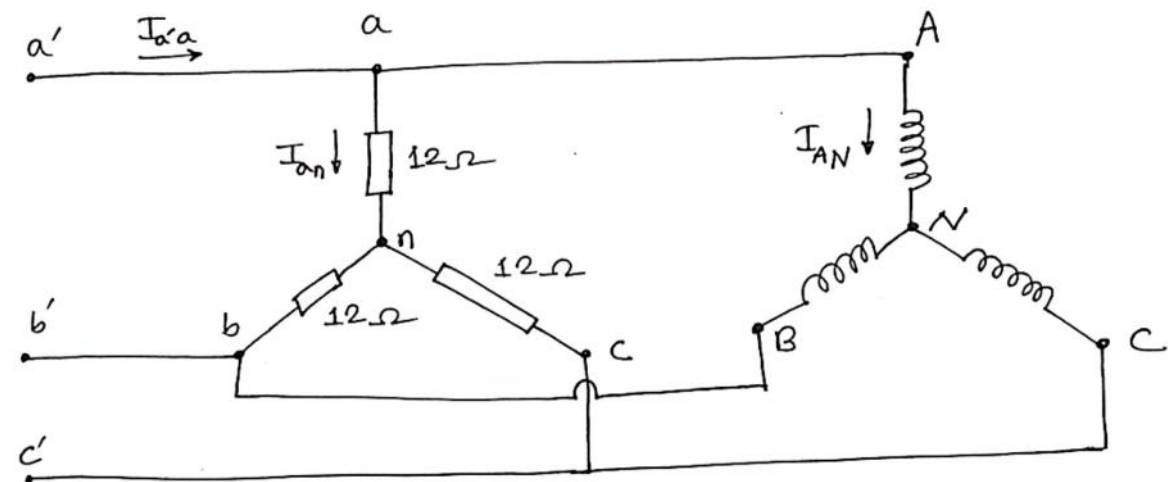
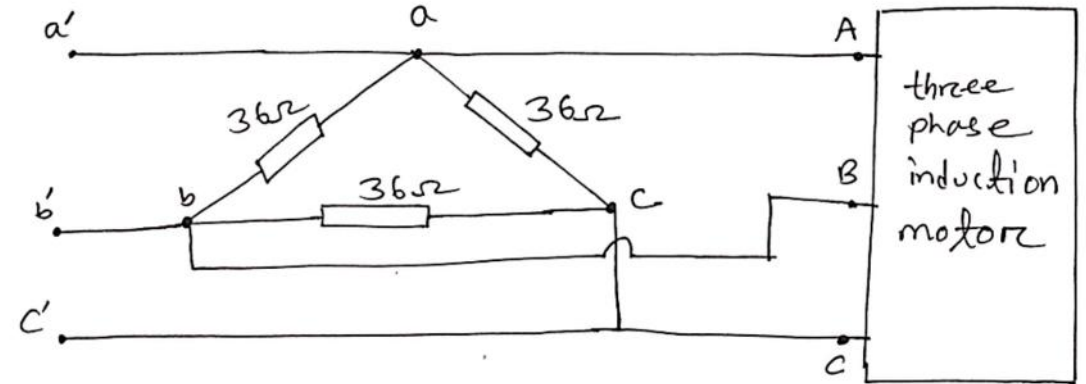
Input line current of motor,

$$I_L = \frac{4337.2}{\sqrt{3} V_L \cos \theta_M} = \frac{4337.2}{\sqrt{3} \times 220 \times 0.866} = 13.14 \text{ A}$$

phase angle of motor, $\theta_M = \cos^{-1}(0.866) = 30^\circ$

Motor phase and line current in complex form:

$$I_{AN} = 13.14\angle(-30 - 30)^\circ \text{A} = 13.14\angle -60^\circ \text{A}$$



Balanced Polyphase Circuits

Resistance furnace current,

$$I_{an} = V_{an}/12 = 127\angle-30^\circ/12 = 10.58\angle-30^\circ \text{ A}$$

The line Current,

$$I_{a'a} = I_{an} + I_{AN} = 10.58\angle-30^\circ + 13.14\angle-60^\circ = 22.92\angle-46.6^\circ \text{ A}$$

Power factor,

$$\cos(-30^\circ + 46.6^\circ) = 0.958 \text{ lagging}$$

kilovolt-amperes,

$$\sqrt{3} V_L I_L = \sqrt{3} \times 220 \times 22.92 = 8733.7 \text{ VA}$$

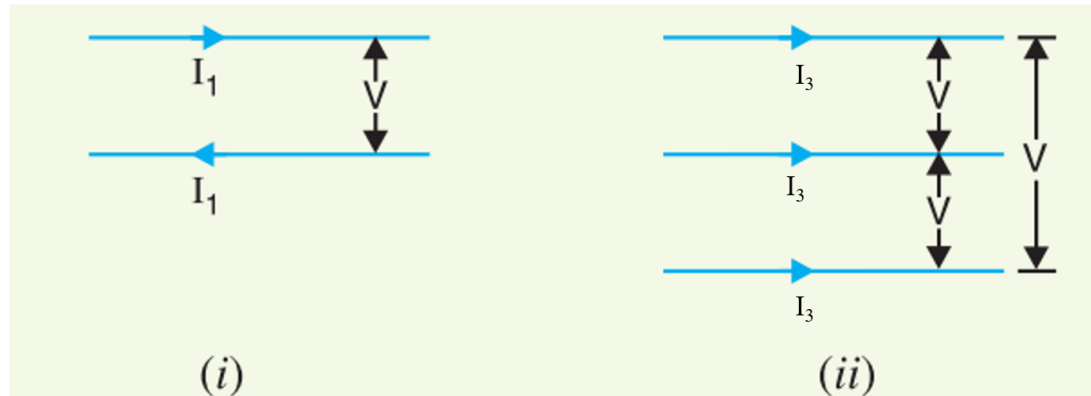


Balanced Polyphase Circuits

Copper ratio of three phase and single phase system under fixed condition:

"Under Fixed Conditions" classically means the following four parameters are held identical for both the three-phase and single-phase systems:

1. Same Amount of Power Transmitted (P)
2. Same Line Voltage (V): (for single-phase, this is its only voltage; for three-phase, this is the line-to-line voltage)
3. Same Distance / Length of Line (l)
4. Same Power Loss
5. Same Power Factor



Power for single phase, $P_1 = V I_1 \cos \theta$

Power for three phase, $P_3 = \sqrt{3} V I_3 \cos \theta$

Since, $P_1 = P_3$

$$\Rightarrow V I_1 \cos \theta = \sqrt{3} V I_3 \cos \theta$$

$$\Rightarrow \boxed{I_1 = \sqrt{3} I_3}$$

$$(\text{Power loss})_1 = (\text{Power loss})_3$$

$$\Rightarrow 2 I_1^2 R_1 = 3 \times I_3^2 R_3$$

$$\Rightarrow \frac{R_1}{R_3} = \frac{3}{2} \left(\frac{I_3}{I_1} \right)^2 = \frac{3}{2} \left(\frac{I_3}{\sqrt{3} I_3} \right)^2$$

$$\Rightarrow \frac{R_1}{R_3} = \frac{1}{2}$$

$$\Rightarrow \frac{(P l / a_1)}{(P l / a_3)} = \frac{1}{2} \Rightarrow \boxed{\frac{a_3}{a_1} = \frac{1}{2}}$$

$$\frac{\text{Copper of three phase}}{\text{Copper of single phase}} = \frac{3 a_3 l \rho}{2 a_1 l \rho} = \frac{3}{2} \cdot \left(\frac{a_3}{a_1} \right) = \frac{3}{2} \cdot \frac{1}{2} = \frac{3}{4}$$

The above shows that the same amount of power may be transmitted a fixed distance with a fixed line loss with only three-fourths of the amount of copper that would be required for single-phase.

Balanced Polyphase Circuits



Historical:

Nikola Tesla (1856–1943) was a Croatian-American engineer whose inventions—among them the induction motor and the first polyphase ac power system—greatly influenced the settlement of the ac versus dc debate in favor of ac. He was also responsible for the adoption of 60 Hz as the standard for ac power systems in the United States.

Born in Austria-Hungary (now Croatia), to a clergyman, Tesla had an incredible memory and a keen affinity for mathematics. He moved to the United States in 1884 and first worked for Thomas Edison. At that time, the country was in the “battle of the currents” with George Westinghouse (1846–1914) promoting ac and Thomas Edison rigidly leading the dc forces. Tesla left Edison and joined Westinghouse because of his interest in ac. Through Westinghouse, Tesla gained the reputation and acceptance of his polyphase ac generation, transmission, and distribution system. He held 700 patents in his lifetime. His other inventions include high-voltage apparatus (the tesla coil) and a wireless transmission system. The unit of magnetic flux density, the Tesla, was named in honor of him.



Nikola Tesla (1856–1943)

Balanced Polyphase Circuits



Enhancing Your Skills and Your Career:

- Developing and enhancing your “ability to identify, formulate, and solve engineering problems” is a primary focus of textbook.
- “An understanding of professional and ethical responsibility” is required of every engineer. To some extent, this understanding is very personal for each of us.
- Your responsibility both professionally and ethically is to perform in a manner that does not harm those around you and to whom you are responsible. Clearly, developing this capability will take time and maturity on your part. I recommend practicing this by looking for professional and ethical components in your day-to-day activities.

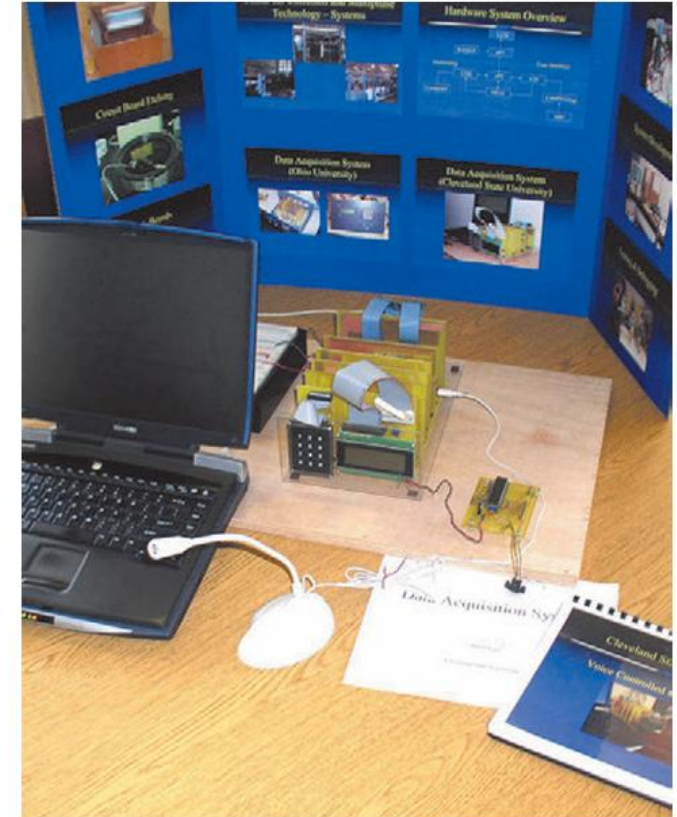


Photo by Charles Alexander

Unbalanced Polyphase Circuits



Any polyphase load in which the impedance in one or more phases differs from those of other phases is said to be unbalanced. Even though the load impedances of the various phases are identical, one of the methods of calculating unbalanced loads must be employed if the voltages impressed on the load are unequal and differ in phase by angles which are not equal.

Unbalanced Polyphase Circuits



Example 1. Given the unbalanced delta load shown in Fig. 1. Calculate all currents for the three-phase balanced voltages shown on the figure, if the voltage sequence is ab–ca–bc.

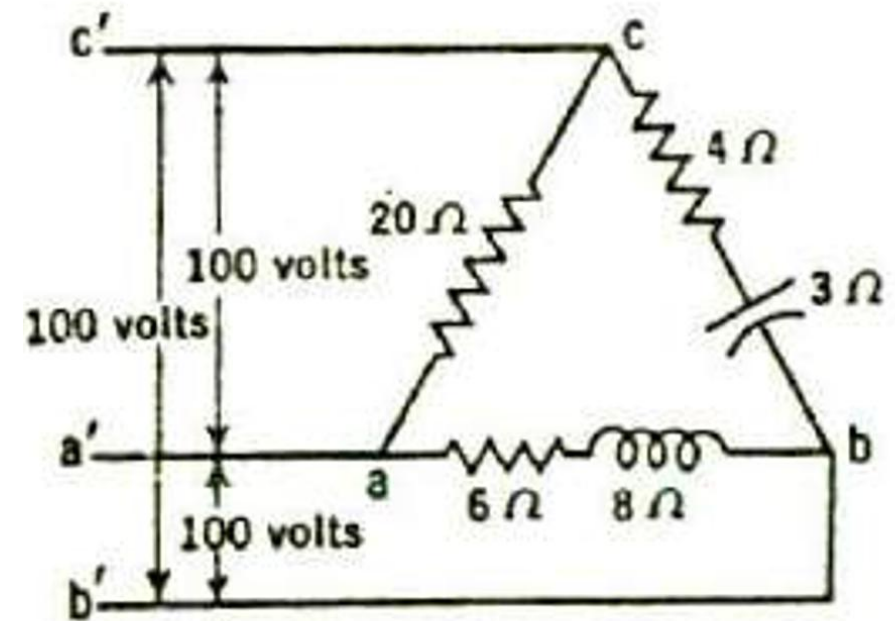


FIG. 1. Unbalanced delta load.
See example 1.

Unbalanced Polyphase Circuits



Example 2. A balanced set of three-phase voltages is connected to an unbalanced set of wye-connected impedances as shown in Fig. 2. The following values are assumed to be known:

$$V_{ab} = 212 \angle 90^\circ \text{ volts}$$

$$V_{bc} = 212 \angle -150^\circ \text{ volts}$$

$$V_{ca} = 212 \angle -30^\circ \text{ volts}$$

$$Z_{an} = 10 + j0 \text{ ohms}$$

$$Z_{bn} = 10 + j10 \text{ ohms}$$

$$Z_{cn} = 0 - j20 \text{ ohms}$$

The line currents $I_{a'a}$, $I_{b'b}$, and $I_{c'c}$ are to be determined by the wye to delta conversion method.

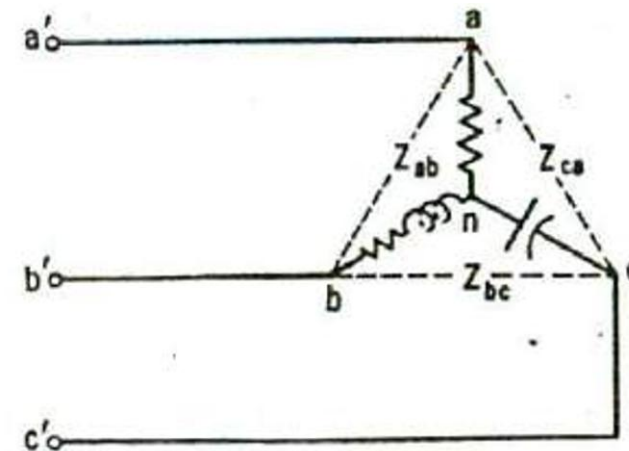


FIG. 2. Conversion from a wye-connected load to an equivalent delta-connected load.

Unbalanced Polyphase Circuits



Example 3: The generated voltages and impedances for Fig. 5 are given as follows:

$$E_{an} = 1000\angle 0^\circ, E_{bn} = 1000\angle -120^\circ, E_{cn} = 1000\angle -240^\circ,$$

$$Z_{na} = 2+j8, Z_{aa'} = 1+j2, Z_{a'n'} = 19+j18,$$

$$Z_{nb} = 2+j8, Z_{bb'} = 1+j2, Z_{b'n'} = 49-j2,$$

$$Z_{nc} = 2+j8, Z_{aa'} = 1+j2 \text{ and } Z_{c'n'} = 29+j50.$$

Find the line currents, phase voltages, and line voltages at the load end.

$$Z_a = 2+j8+1+j2+19+j18 = 22+j28 \Omega$$

$$Z_b = 2+j8+1+j2+49-j2 = 52+j8 \Omega$$

$$Z_c = 2+j8+1+j2+29+j50 = 32+j60 \Omega$$

Using mesh analysis,

$$(Z_a + Z_b)I_1 - Z_b I_2 = E_{an} - E_{bn}$$

$$\Rightarrow (22+j28 + 52+j8) I_1 - (52+j8) I_2 = 1000\angle 0^\circ - 1000\angle -120^\circ$$

$$\Rightarrow (74+j36) I_1 - (52+j8) I_2 = 1500+j866 \dots\dots(i)$$

$$(Z_b + Z_c)I_2 - Z_b I_1 = E_{bn} - E_{cn}$$

$$\Rightarrow (52+j8 + 32+j60) I_2 - (52+j8) I_1 = 1000\angle -120^\circ - 1000\angle -240^\circ$$

$$\Rightarrow -(52+j8) I_1 + (84+j68) I_2 = -j1732 \dots\dots(ii)$$

Solving equation (i) & (ii),

$$I_1 = 16\angle -34.9^\circ \text{ A and } I_2 = 20.7\angle -109.2^\circ \text{ A}$$

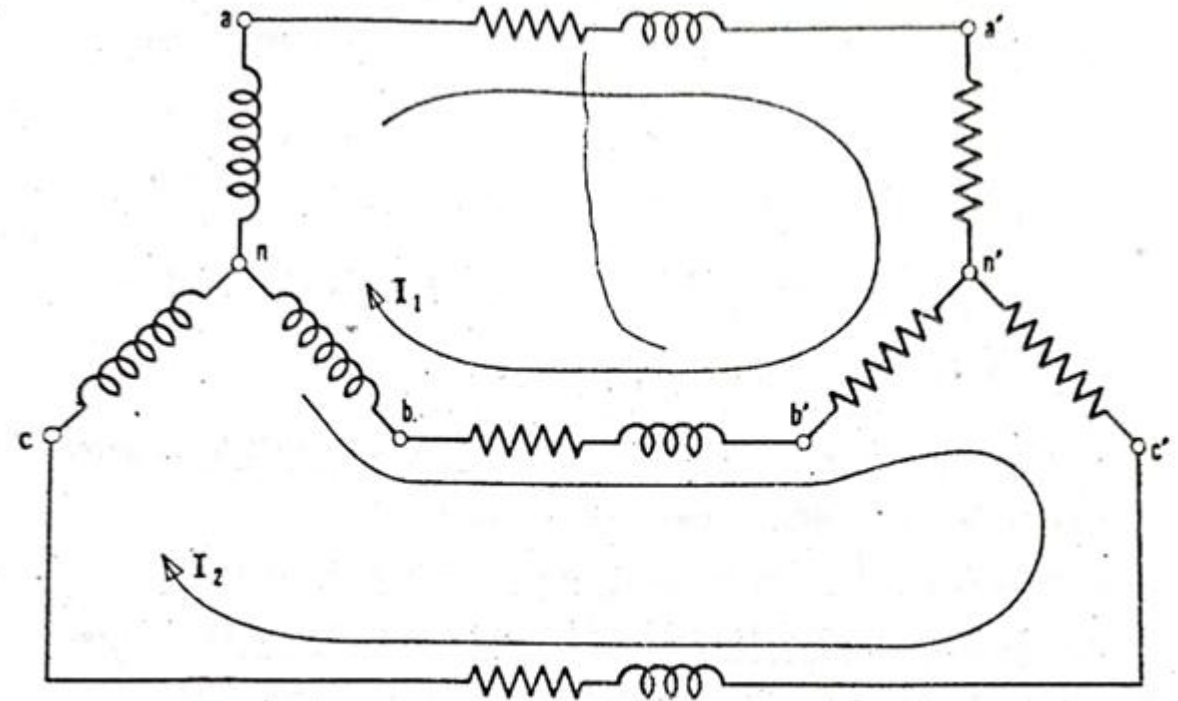


Fig. 5.

Unbalanced Polyphase Circuits



Line currents:

$$I_{aa'} = I_1 = 16 \angle -34.9^\circ \text{ A}$$

$$I_{cc'} = -I_2 = -20.7 \angle -109.2^\circ \text{ A} = 20.7 \angle 70.8^\circ \text{ A}$$

$$I_{bb'} = -(I_{aa'} + I_{cc'}) = 22.47 \angle -152.4^\circ \text{ A}$$

Phase voltages at load end :

$$V_{a'n'} = 16 \angle -34.9^\circ \times (19 + j18) = 418.76 \angle 8.5^\circ \text{ V}$$

$$V_{b'n'} = 22.47 \angle -152.4^\circ \times (49 - j2) = 1102.32 \angle -154.8^\circ \text{ V}$$

$$V_{c'n'} = 20.7 \angle 70.8^\circ \times (29 + j50) = 1196.48 \angle 130.7^\circ \text{ V}$$

Line voltages at load end :

$$V_{a'b'} = V_{a'n'} - V_{b'n'} = 418.76 \angle 8.5^\circ - 1102.32 \angle -154.8^\circ = 1508.2 \angle 20.6^\circ \text{ V}$$

$$V_{b'c'} = V_{b'n'} - V_{c'n'} = 1102.32 \angle -154.8^\circ - 1196.48 \angle 130.7^\circ = 1393.46 \angle -98.9^\circ \text{ V}$$

$$V_{c'a'} = V_{c'n'} - V_{a'n'} = 1196.48 \angle 130.7^\circ - 418.76 \angle 8.5^\circ = 1463.1 \angle 144^\circ \text{ V}$$

Unbalanced Polyphase Circuits



The Y-Y System with Neutral Connection:

Line currents:

$$I_{a'a} = \frac{E_{n'a'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{an})}; \quad I_{b'b} = \frac{E_{n'b'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{bn})};$$

$$I_{c'c} = \frac{E_{n'c'} - I_{nn'}Z_n}{(Z_g + Z_l + Z_{cn})}$$

Let,

$$Z_g + Z_l + Z_{an} = Z_a$$

$$Z_g + Z_l + Z_{bn} = Z_b$$

$$Z_g + Z_l + Z_{cn} = Z_c$$

Applying KCL at neutral point,

$$I_{a'a} + I_{b'b} + I_{c'c} = I_{nn'}$$

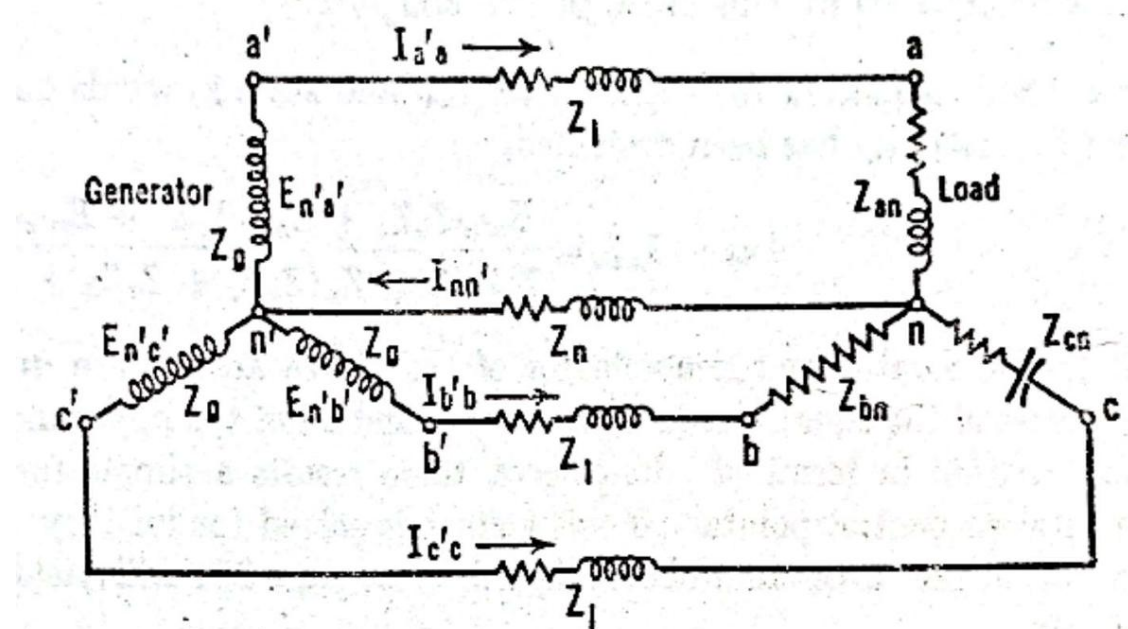


FIG. 8. A four-wire three-phase system.

Unbalanced Polyphase Circuits



$$I_{a'a} + I_{b'b} + I_{c'c} = I_{nn'}$$

$$\Rightarrow \frac{E_{n'a'} - I_{nn'}Z_n}{Z_a} + \frac{E_{n'b'} - I_{nn'}Z_n}{Z_b} + \frac{E_{n'c'} - I_{nn'}Z_n}{Z_c} = I_{nn'}$$

$$\Rightarrow \frac{E_{n'a'}Z_bZ_c - I_{nn'}Z_nZ_bZ_c + E_{n'b'}Z_aZ_c - I_{nn'}Z_aZ_c + E_{n'c'}Z_aZ_b - I_{nn'}Z_nZ_aZ_b}{Z_aZ_bZ_c} = I_{nn'}$$

$$\Rightarrow E_{n'a'}Z_bZ_c + E_{n'b'}Z_aZ_c + E_{n'c'}Z_aZ_b = I_{nn'}[Z_aZ_bZ_c + Z_n(Z_bZ_c + Z_cZ_a + Z_aZ_b)]$$

$$I_{nn'} = \frac{E_{n'a'}Z_bZ_c + E_{n'b'}Z_cZ_a + E_{n'c'}Z_aZ_b}{Z_aZ_bZ_c + Z_n(Z_bZ_c + Z_cZ_a + Z_aZ_b)}$$

Unbalanced Polyphase Circuits

Methods of Checking voltage phase sequence :



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



Unbalanced Polyphase Circuits



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